

GarageBand VR Drums

Week 9 Testing Plan

GarageBand VR Drums is an interactive virtual drumming experience that allows users to play a drum kit, record a short performance, replay it, and save their performance through an interactive control panel. Instead of interacting with a traditional music application, users are placed in a virtual performance environment where drumming is performed through spatial interaction with virtual drumsticks. The prototype aims to make the basic drumming and recording experience engaging, immersive, and easy to understand through a simple, game-like interaction flow.

Following the Week 5 usability testing and evaluation, the prototype was further developed and refined based on issues identified during the previous testing session. The Week 9 testing will therefore focus on evaluating the revised interaction flow and determining whether the changes made after the previous evaluation have improved the clarity and usability of the experience.

Testing Objective

The Week 5 evaluation identified several areas where participants experienced confusion, particularly around discovering the controls, understanding different recording and playback states, and controlling the drumsticks. Based on these findings, the prototype was iterated to provide a clearer overall interaction flow and more guidance and feedback during the experience.

This test aims to evaluate whether users who were not involved in developing the prototype can now move through the full experience of choosing a musician, starting a recording, playing the drums, stopping the recording, playing it back, and saving it without unnecessary confusion or assistance. It will also identify whether any issues from the previous evaluation remain and whether new usability problems have appeared after the changes.

Testing Methodologies

This testing plan will use task-based usability testing, observation, and think-aloud to evaluate the revised Unity prototype. Participants will be given goals to complete without being told exactly which controls to use, while being encouraged to explain what they are thinking, what they expect to happen, and anything they find confusing. The results will also be compared with observations from the Week 5 evaluation to determine whether previously identified usability issues have been improved.

Prototype description/requirements

The prototype has been further developed following the Week 5 evaluation to provide a more complete and understandable interaction flow. Users can choose a musician, enter the drum experience, start recording, play a short drum performance, stop the recording, listen to the playback, and save the recording. The prototype also provides visual and audio feedback to communicate different interaction states and includes a Help panel that participants can access if they need additional help. These elements allow the testing to evaluate whether the revised prototype has improved the clarity of the experience and made it easier for users to understand what is happening and what they should do next.

Data collection method

During the testing process, I will observe each participant while they complete the tasks and record whether they complete each step independently, hesitate, perform an unexpected action, or require assistance. I will also take notes on participant comments during the think-aloud process, any unexpected prototype behaviour or bugs, and feedback given after the tasks are completed.

Testing Setup

Before each test, I will open the Unity prototype at the starting screen and check that the musician selection, drum controls, audio, recording, playback, and save functions are working correctly. I will also reset any previous recording state and prepare a note sheet for observations. Before beginning the tasks, I will briefly explain that the prototype is being tested rather than the participant's ability. I will point out the help button in the top-left corner, which participants can use if they need instructions during the test, and explain how to close the dialogue panels.

I will also briefly guide participants on how to control the drumsticks using the cursor, as this interaction is less intuitive when testing the prototype directly in Unity compared to using the intended VR controllers. Other than this basic control guidance, I will avoid demonstrating how to complete the tasks unless the participant becomes completely stuck.

Testing process

Introduction: Thank you for testing my GarageBand VR Drums prototype. During this session, I will give you a series of tasks to complete using the prototype. Please try to complete each task on your own first so I can see whether the flow and controls are clear to someone using it for the first time. While completing the tasks, please think aloud and tell me what you are looking for, what you expect to happen, or anything that feels confusing. If you become completely stuck, you can ask me for help. Before we begin, you can open and read the How To Play panel using the Help button. I will also briefly explain how to close the dialogue panel and how the drumstick movement works, as controlling the drumsticks with a cursor in the Unity version is slightly different from how they would be controlled in VR. Is it okay if I record this testing session for observation and evaluation purposes? **(Approximately 1 minute)**

1. **Task 1 - Choose a musician:** "Choose a musician you would like to play as and continue into the experience." Observe whether the selection and next step are clear. **(Approximately 30 seconds)**
2. **Task 2 - Start recording:** "You want to record your drum performance. Start a new recording." Observe whether the participant can identify the correct control and recognise that recording has started. **(Approximately 30 seconds)**
3. **Task 3 - Play drums:** "Play a short drum performance." Observe whether the drum controls are understandable and whether anything unexpected happens during recording. **(Approximately 1 minute)**
4. **Task 4 - Stop recording:** "You have finished your performance. End the recording." Observe whether the participant understands how to stop and whether the change of state is clear. **(Approximately 30 seconds)**
5. **Task 5 - Playback:** "You want to listen to the performance you just recorded." Observe whether the participant can identify the playback control and understand what is happening. **(Approximately 30 seconds)**

6. **Task 6 - Save:** “You are happy with your performance and want to keep it.” Observe whether the participant can identify how to save and whether the save confirmation is clear. (**Approximately 30 seconds**)
7. **Post-test questions:** “Was there any point where you were unsure what to do next?”, “Did anything happen differently from what you expected?”, and “What would you change to make the experience clearer?” (**Approximately 3 - 5 minutes**)

Total estimated testing time: approximately 8 - 10 minutes per participant.